

Covariance & Correlation:-

Covariance & Correlation are two statistical measurements used to determine the relationship between two variables. Both are used to understand how changes in one variable are associated with changes in another variable.

Covariance:-

Covariance is a measure of how much two random variables change together. If the variables tend to increase and decrease together, the covariance is positive. If one tends to increase when the other decreases the covariance is negative.

X	Y
-2	-3
-4	-5
6	7
8	9

$x \uparrow y \uparrow$
 $x \downarrow y \downarrow$ +ve

X	Y
7	10
6	12
5	14
4	16

$x \downarrow y \uparrow$
 $x \uparrow y \downarrow$ -ve

Single of Hour

Price

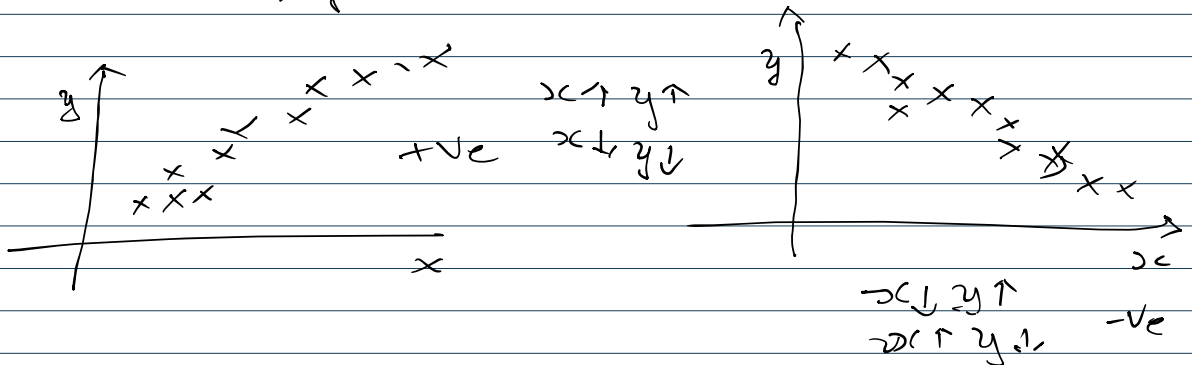
1700 sqft

45 lakhs

1500 sqft

50 lakhs

$x \uparrow \rightarrow y \uparrow$



Covariance

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\text{Cov}(x, x) = \frac{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})}{n-1} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Terms

Terms

x_i = Data point random variable (x)

$\bar{x}$ $\Rightarrow$ Sample of x

y_i $\Rightarrow$ Data point random variable (y)

$\bar{y}$ $\Rightarrow$ Sample mean of y

$$\boxed{\therefore \text{Cov}(x, x) = \text{Var}(x)}$$

Students

Hours Studied (x) / week

Exam Score (y)

2	50
3	60
4	70
5	80
6	90

$x \uparrow y \uparrow$

$x \downarrow y \downarrow$ +ve Covariance

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\bar{x} = \frac{2+3+4+5+6}{5} = \frac{20}{5} = 4$$

$$\bar{y} = \frac{50+60+70+80+90}{5} = \frac{350}{5} = 70$$

2	50
3	60
4	70
5	80
6	90

$$\begin{aligned} \text{Cov}(x, y) &= (2-4)(50-70) + (3-4)(60-70) + (4-4)(70-70) \\ &\quad + (5-4)(80-70) + (6-4)(90-70) \\ &= (-2)(-20) + (-1)(-10) + 0 + (1)(10) + (2)(20) \\ &= 100 = 20 \end{aligned}$$

$\text{Cov}(x, y) = +20 \rightarrow +ve \text{ Covariance}$

Advantages

Disadvantages

① Identifies the Relationship b/w x & y

① Covariance doesn't have a specific limit

Correlation $\rightarrow$ Pearson correlation coefficient
 $\rightarrow$ Spearman correlation coefficient

-1 to 1

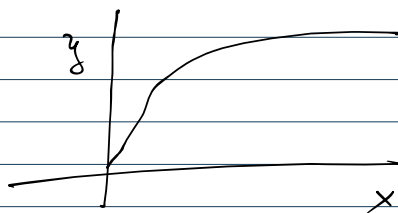
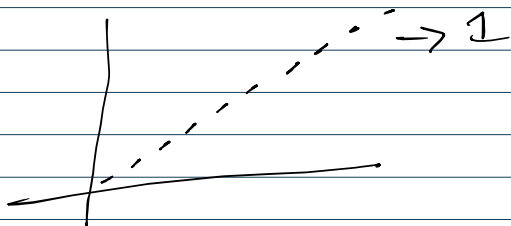
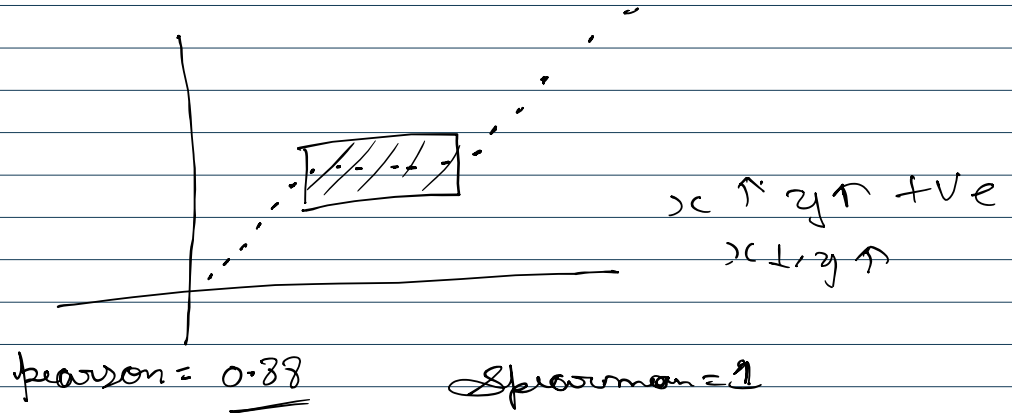
→ Spearman correlation coefficient

i) Pearson Correlation Coefficient: [-1 to 1]

$$r_{x,y} = \frac{\text{Cov}(x,y)}{\sigma_x \cdot \sigma_y}$$

(x) +1 → +ve Correlation $x \uparrow y \uparrow$
 (x) -1 → -ve Correlation $x \uparrow y \downarrow$

ii) Spearman Rank Correlation



$$r_s = \frac{\text{Cov}(R(x), R(y))}{\sigma(R(x)) \cdot \sigma(R(y))}$$

$x \quad y \quad R(x) \quad R(y)$

1		1	1
3	2	2	2
5	4	3	3
7	6	4	5
9	8	5	4
	7		

Feature Selection

Discrete

System

large $\leftrightarrow$ cold

Mild fever

$\rightarrow$ Covid-19